

AI ASSISTED CODING

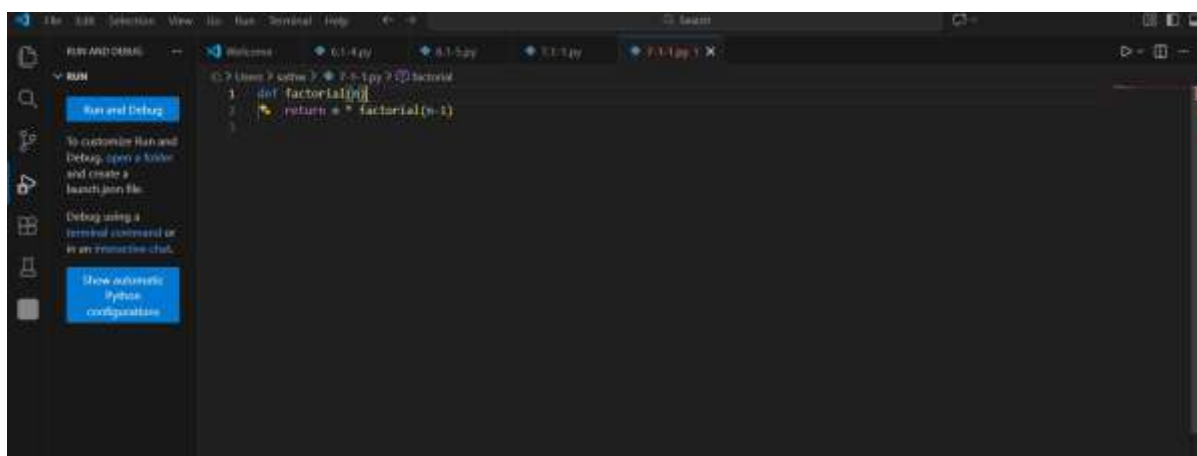
Assignment-7.1

HTNO:2303A510F1

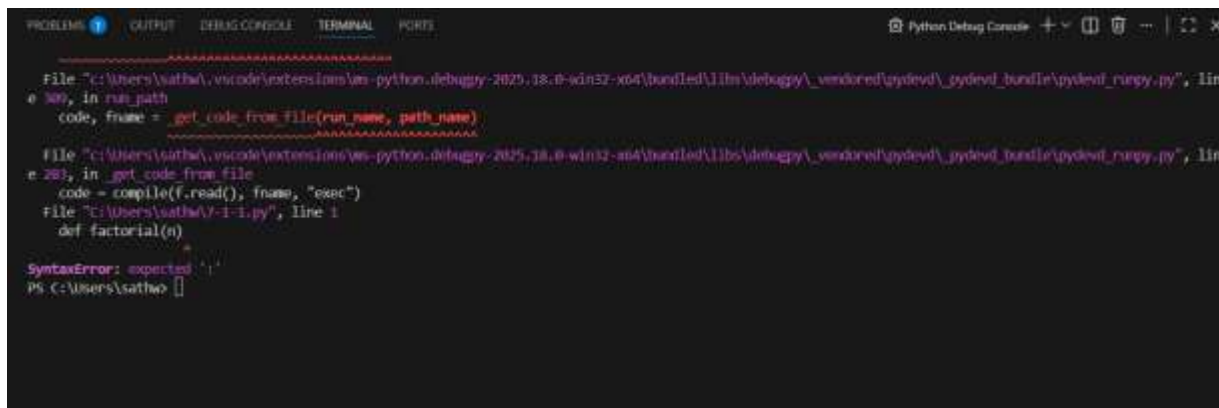
Batch:14

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Task 1: Debugging a Recursive Calculation Module



Output:



After Debugging

I found the syntax error! You're missing a **colon (:)** at the end of the function definition line.

In Python, all function definitions must end with a colon to indicate the start of the function body. Here's the corrected code:

```
def factorial(n):  
    return n * factorial(n-1)
```

Key points:

- **The colon (:)** signals to Python that an indented block follows
- The indentation of the [return](#) statement shows it's part of the function body

Additional observation: This recursive function has an issue—it will cause infinite recursion because it never has a **base case** (a condition to stop recursing). You'd want to add a base case:

```
def factorial(n):  
    if n <= 1:  
        return 1  
    return n * factorial(n-1)
```

This prevents the recursion from continuing indefinitely by returning 1 when [n](#) reaches 1 or less.

```
C:\Users\sathw> 7.1-2.py  
1 # Task 1: Debugging a Recursive Calculation Module  
2 # Factorial using recursion  
3  
4 def factorial(n):  
5     if n < 0:  
6         return "Factorial is not defined for negative numbers"  
7     elif n == 0 or n == 1:  
8         return 1  
9     else:  
10        return n * factorial(n - 1)  
11  
12  
13 number = int(input("Enter a number: "))  
14 result = factorial(number)  
15  
16 print(f"Factorial of {number} is {result}")  
17
```

Output:

```
Python debugpy - 2023.18.0-win32-x64 (bundled) (1105 debugpy \la  
Enter a number: 5  
Factorial of 5 is 120  
PS C:\Users\sathw>
```

Task – 2:

Task 2: Fixing Data Type Errors in a Sorting Utility

```
C:\Users\sathw > 7.1.2.py > ...
1 values = [10, "20", 5, "15"] values = [10, "20", 5, "15"]
2 values.sort()
3 print(values)
4
```

Output:

```
PS C:\Users\sathw >
PS C:\Users\sathw >
PS C:\Users\sathw > c:\cd 'c:\Users\sathw'; & 'c:\Users\sathw\AppData\Local\Microsoft\WindowsApps\python1.13.exe' 'c:\Users\sathw\.vscode\extensions\ms-p
ython.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '69153' '-' 'c:\Users\sathw\7.1.2.py'
```

After Debugging:

The code has a **bug**. Here's what's happening:

```
data = [10, "5", 3, "20", 1]
```

```
print(sorted(data))
```

The Problem:

You're trying to sort a list that contains **both integers and strings**. In Python 3, you cannot compare integers with strings directly, so `sorted()` fails with a `TypeError`.

Why it fails:

- The list has: 10 (int), "5" (string), 3 (int), "20" (string), 1 (int)
- When sorting, Python tries to compare 10 with "5", but integers and strings aren't comparable

```
C:\Users\sathw > 7.1.2.py > ...
1 values = [10, "20", 5, "15"]
2 cleaned_values = [int(x) for x in values]
3 cleaned_values.sort()
4
5 print("\nTask 2 Output:")
6 print("Sorted values:", cleaned_values)
7
```

Output:

```
PS C:\Users\sathw > c:\cd 'c:\Users\sathw'; & 'c:\Users\sathw\AppData\Local\Microsoft\WindowsApps\python1.13.exe' 'c:\Users\sathw\.vscode\extensions\ms-p
ython.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '60165' '-' 'c:\Users\sathw\7.1.2.1.py'
Sorted data: [1, 3, 5, 10, 20]
PS C:\Users\sathw >
```

Task - 3:

Task 3: Improving File Handling Reliability

```
C:\Users\sathw> python 7.13.py -
1 # Task 3: Improving file handling reliability
2
3 try:
4     with open("data.txt", "r") as file:
5         content = file.read()
6         print(content)
7 except FileNotFoundError:
8     print("Error: The file 'data.txt' does not exist.")
9 except PermissionError:
10    print("Error: Permission denied to read the file 'data.txt'.")
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\sathw> & 'c:\Users\sathw\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\sathw\.vscode\extensions\ms-python.de
buggy-2025.12.8-win32-x64\handed\lib\debugpy\launcher' '51200' '-' 'c:\Users\sathw\7.13.py'
Error: The file 'data.txt' does not exist.
PS C:\Users\sathw>
```

After Debugging:

What it does:

1. **open("data.txt", "r")** - Opens a file named data.txt in **read mode** ("r")
2. [file.read\(\)](#) - Reads all the content from the file
3. [print\(content\)](#) - Displays the file content

Why

it's

failing:

The file data.txt doesn't exist in your working directory, so Python throws a FileNotFoundError.

```
C:\> Users > sathw > 7.1.3.py > ...
1 # Task 3: Improving File Handling Reliability
2
3 # Create sample data file
4 try:
5     with open("data.txt", "w") as file:
6         file.write(
7             "Hello, World!\n"
8             "This is sample data.\n"
9             "File handling in Python."
10        )
11 except Exception as e:
12     print(f"Error creating file: {e}")
13
14 # Read and display the file
15 try:
16     with open("data.txt", "r") as file:
17         content = file.read()
18         print("File Content:")
19         print(content)
20 except FileNotFoundError:
21     print("Error: data.txt file not found!")
22
```

Output:

```
\extensions\vs-python-debugpy-2025.10.0-win32-x64\bundle\libs\debugpy\launcher "55028" -- "C:\Users\sathw\7.1.3.py"
File Content:
Hello, World!
This is sample data.
File handling in Python.
PS C:\Users\sathw>
```

Task – 4:

Task 4: Handling Runtime Errors Gracefully in Loops

```
C:\> Users > sathw > 7.1-4.py > _
1 # Task 4: Handling Runtime Errors Gracefully in Loops
2
3 values = [10, 5, 0, 2]
4
5 for v in values:
6     try:
7         print(10 / v)
8     except ZeroDivisionError:
9         print("Error: Division by zero skipped")
10
```

Output :

```

PS C:\Users\sathw> c:: cd 'c:\Users\sathw'; & 'c:\Users\sathw\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\sathw\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '52896' '--' 'C:\Users\sathw\7.1-4.py'
1.0
2.0
Error: Division by zero skipped
5.0
PS C:\Users\sathw>

```

After Debugging:

What it does:

- Loops through each value in the list: 10, 5, 0, 2
- Divides 10 by each value and prints the result

Why

it's

failing:

When v = 0, Python tries to execute 10 / 0, which causes a **ZeroDivisionError** (you cannot divide by zero).

```

C:\Users\sathw> 7.4-5.py > ...
1 # Task 4: Handling Runtime Errors Gracefully in Loops
2
3 values = [10, 5, 0, 2]
4
5 for v in values:
6     try:
7         result = 10 / v
8         print(result)
9     except ZeroDivisionError:
10        print(f"Error: Cannot divide by {v}")
11

```

Output:

```

PS C:\Users\sathw> c:: cd 'c:\Users\sathw'; & 'c:\Users\sathw\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\sathw\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '51378' '--' 'C:\Users\sathw\7.4-5.py'
1.0
2.0
Error: Cannot divide by 0
5.0
PS C:\Users\sathw>

```

Task -5:

Task 5: Debugging Class Initialization Errors

```

C:\Users\sathw> 7.1-5.py > ...
1 class Student:
2     def __init__(self, name, age): # added self
3         self.name = name
4         self.age = age
5
6     def display(self): # added self
7         print(self.name, self.age)
8
9
10 s = Student("Sathwik", 21)
11 s.display()
12
13

```

Output :

```
PS C:\Users\sathw> cd 'C:\Users\sathw'; & 'C:\Users\sathw\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'C:\Users\sathw\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-and-hurd\lib\debugpy\launcher' 'sathw' '-' 'C:\Users\sathw\7.1-5.py'
name: Sathwik reddy
Age: 21
PS C:\Users\sathw>
PS C:\Users\sathw>
```

After debugging:

What it's trying to do:

- Create a [Student](#) class with name and age attributes
- Initialize a student object and display their details

Why it's failing:

1. [__init__](#) method: Missing [self](#) parameter
 - Should be: [def __init__\(self, name, age\):](#)
2. [display\(\)](#) method: Missing [self](#) parameter AND references to variables
 - Should be: [def display\(self\):](#)
 - Should reference: [self.name](#) and [self.age](#)

```
C:\Users\sathw> 7.1-5.py > ...
1 class Student:
2     def __init__(self, name, age): # added self
3         self.name = name
4         self.age = age
5
6     def display(self): # added self
7         print(self.name, self.age)
8
9
10 s = Student("Sathwik reddy", 21)
11 s.display()
12
13
```

Output:

```
PS C:\Users\sathw> cd 'C:\Users\sathw'; & 'C:\Users\sathw\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'C:\Users\sathw\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-and-hurd\lib\debugpy\launcher' 'sathw' '-' 'C:\Users\sathw\7.1-5.py'
name: Sathwik reddy
Age: 21
PS C:\Users\sathw>
PS C:\Users\sathw>
```